Myocardial Infarction Risk Prediction Using Machine Learning

UE24CS352A Machine Learning | Mini-Project Write-up | PES University, Bengaluru

Team	Team <No.> - Gaurav S. (PES1UG24AM104) and <Teammate name (USN)>	**Problem No.**	<Serial no. as per master list>

1. Problem Statement

Cardiovascular disease, and myocardial infarction (MI) in particular, is a leading cause of death, and early identification of high-risk patients allows timely intervention. The goal of this project is to build a machine learning system that, given 13 routine clinical measurements of a patient, **predicts whether the patient is at higher or lower risk** of heart disease / MI and reports the model's estimated probability. We also explore the structure of the patient data with unsupervised learning (PCA, K-Means, DBSCAN).

2. Dataset

We used the *Heart Attack Analysis & Prediction* (heart disease) dataset: **1,025 records x 14 columns**. It contained **723 duplicate rows**; after removal **302 unique patients** remain, with no missing values. The binary target is balanced (164 higher-risk = 54.3%, 138 lower-risk = 45.7%). Features: age, sex, chest-pain type (cp), resting blood pressure (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), maximum heart rate (thalach), exercise-induced angina (exang), ST depression (oldpeak), ST slope, number of major vessels (ca) and thalassemia result (thal). EDA showed the strongest correlations with the target for exang (-0.44), oldpeak (-0.43), cp (+0.43), thalach (+0.42) and ca (-0.41).

3. Approach

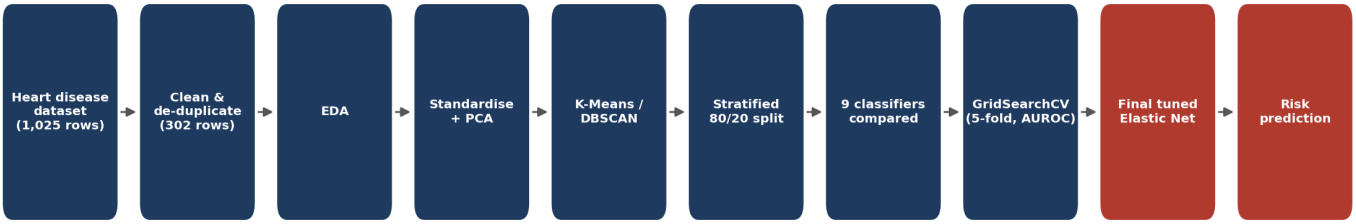


Figure 1. End-to-end pipeline.

- **Pre-processing:** duplicate removal, missing-value check, standardisation (`StandardScaler`; fitted on the training set only for supervised models to avoid leakage).
- **Unsupervised analysis:** PCA to 2 components, then K-Means (k = 2-10) and DBSCAN (eps swept 0.2-2.0), scored with Adjusted Rand Index (ARI) against the true label and silhouette score.
- **Supervised models (9):** Logistic Regression, Elastic Net, SVM (linear / RBF / polynomial), Random Forest, Gradient Boosting, Bagging and an MLP neural network (32-16), trained on a stratified 80/20 split (241 train / 61 test, `random_state=42`).
- **Model selection:** accuracy, precision, recall, F1 and AUROC on the hold-out set; `GridSearchCV` (5-fold stratified, AUROC) on the strongest candidates.

4. Implementation Overview

The whole pipeline is implemented in one Python notebook (pandas, NumPy, scikit-learn, matplotlib, seaborn) and hosted in a private GitHub repository with a README. The final tuned model, scaler and feature order are saved with `joblib`. A prediction function, `predict_heart_risk()` / `show_prediction()`, rebuilds the input in the training feature order, applies the saved scaler, and returns the predicted class and the class-1 probability. For example, a 55-year-old male with cp = 1, BP 130, cholesterol 240 and max heart rate 150 is predicted *higher risk* with probability 60.85%.

5. Results

Clustering. PCA retained only 33.2% of the variance in two components. K-Means with k = 2 gave ARI = **0.377** and silhouette = 0.423, so the two clusters partly align with disease status; ARI fell steadily as k increased. DBSCAN (eps = 0.5, min_samples = 5) found one cluster plus 27 noise points (ARI = 0.012): density-based clustering did not separate the classes in 2-D PCA space.

Classification (untuned, 61-patient test set). Table 1 and the ROC curves in Figure 2 compare the nine models.

Model	Acc.	Prec.	Recall	F1	AUROC
Elastic Net	0.803	0.800	0.848	0.824	0.872
Logistic Regression	0.803	0.800	0.848	0.824	0.871
SVM Polynomial	0.820	0.789	0.909	0.845	0.870
Random Forest	0.754	0.765	0.788	0.776	0.870
SVM Linear	0.820	0.806	0.879	0.841	0.869
SVM RBF	0.770	0.771	0.818	0.794	0.842
Bagging	0.770	0.771	0.818	0.794	0.837
Neural Network	0.754	0.765	0.788	0.776	0.832
Gradient Boosting	0.721	0.735	0.758	0.746	0.826

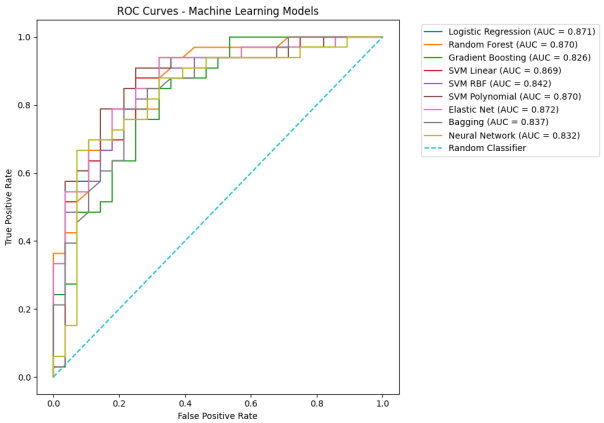


Table 1. Test-set metrics (left); Figure 2. ROC curves of all models (right).

Tuning and final model. Tuned models were compared on the same test set. Elastic Net and Logistic Regression both reached **test AUROC 0.881** (cross-validated AUROC 0.917); tuned SVM-polynomial reached the highest CV AUROC (0.923) but generalised worse

on the test set (AUROC 0.844, precision 0.696), a sign of variance on small data. **Tuned Elastic Net** (C = 0.01, l1_ratio = 0) was selected as the final model: accuracy 0.803, precision 0.784, **recall 0.879**, F1 0.829, AUROC 0.881.

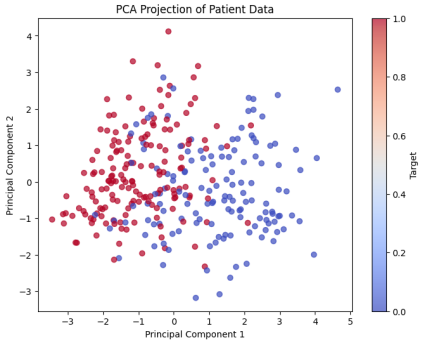


Figure 3. PCA projection coloured by target

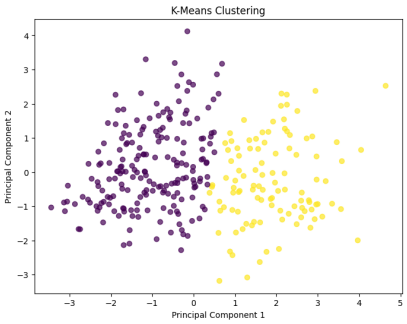


Figure 4. K-Means (k = 2) clusters

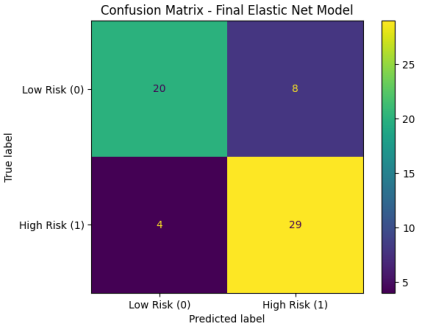


Figure 5. Final model confusion matrix

6. Conclusions

- A complete pipeline (EDA, PCA, clustering, nine classifiers, tuning and a prediction function) was built; the final tuned Elastic Net reaches **AUROC 0.881** and **recall 0.879**, detecting 29 of 33 higher-risk patients (TN 20, FP 8, FN 4, TP 29).
- Simple linear models matched or beat tree ensembles and the neural network on this small tabular dataset; the best models differ by less than the noise expected from a 61-patient test set, so the ranking among them should not be over-interpreted.
- Unsupervised clustering found partial structure (K-Means ARI 0.377) but is not a substitute for supervised prediction.
- **Limitations / future work:** only 302 unique patients and a label that denotes heart-disease presence rather than confirmed MI; repeated or nested cross-validation, probability calibration, feature-importance analysis and external validation are natural next steps. The system is a demonstration and **not** a medical diagnosis tool.